



All images loaded

Docker CLI

An overview about docker cli.

Overview

This guide explains how to install the Data Plane using Docker CLI commands. The Data Plane includes core components like the Gateway and Redis for rate limiting.

Prerequisites

- Install [Docker ↗](#).
- Ensure you have access to [Gravitee Cloud ↗](#), with permissions to install new Gateways.
- Complete the steps in [Prepare your installation](#).

Install Gateway and Configure Redis

To enable API rate-limiting, configure your Gateway to use a rate-limiting repository, such as Redis with the following steps:

1. Create a Docker network with the following command:

```
docker network create gravitee-network
```

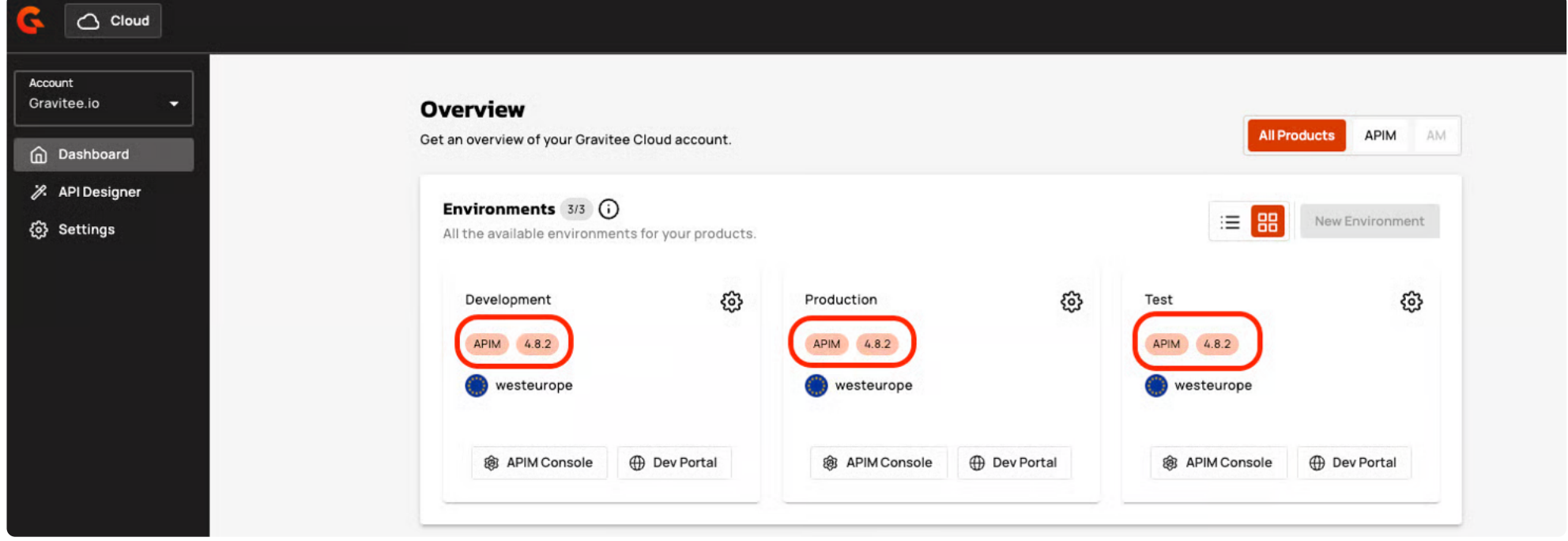
2. Run the command below to start Redis.

```
docker run -d \
  --name gio-apim-hybrid-redis \
  --hostname redis \
  --network gravitee-network \
  -p 6379:6379 \
  redis:7.2-alpine redis-server --requirepass <redis_password>
```

- Replace `<redis_password>` with your own secure password.
3. Run the Gateway with Redis rate limiting enabled with the following command:

```
docker run -d \
  --name gio-apim-hybrid-gateway \
  --hostname apim-gateway \
  --network gravitee-network \
  -p 8082:8082 \
  -e gravitee_ratelimit_type=redis \
  -e gravitee_ratelimit_redis_host=redis \
  -e gravitee_ratelimit_redis_port=6379 \
  -e gravitee_ratelimit_redis_password=<redis_password> \
  -e gravitee_ratelimit_redis_ssl=false \
  -e gravitee_cloud_token=<cloud_token> \
  -e gravitee_license_key=<license_key> \
  graviteeio/apim-gateway:<add_the_gateway_tag> #The gateway version to install. It has to
```

- Replace `<cloud_token>` with your Cloud Token from Gravitee Cloud.
- Replace `<license_key>` with your License Key from Gravitee Cloud.
- Set `redis_password` environment variable: `export redis_password=your_redis_password` or replace `${redis_password}` with your actual Redis password.
- Replace `<add_the_gateway_tag>` with the gateway version to install. It has to align with the control plane of your Gravitee Cloud.



Verification

To confirm that your Hybrid installation is working, complete the following steps:

- [Check container status](#)
- [Check port mapping](#)
- [Ensure the Gateway is listening in your local environment](#)
- [Verify the Redis Connection](#)
- [View the logs](#)
- [Docker CLI](#)

Check container status

- Check the container status using the following command:

```
docker ps
```

The command generates the following output.

CONTAINER ID	IMAGE	COMMAND	CREATED
50016b90785e	graviteeio/apim-gateway:4.11.0	"/bin/gravitee"	About an hour ago
a8d3e6f1c2b4	redis:7.2-alpine	"redis-server --requ..."	About an hour ago

Check port mapping

- To verify the container's port is correctly mapped run the following command:

```
docker port gio-apim-hybrid-gateway
```

The command generates the following output:

```
8082/tcp -> 0.0.0.0:8082
```

Ensure the Gateway is listening in your local environment

1. Open a terminal, and then run the following `curl` command to call the Gateway on its default exposed port:

```
curl -i http://localhost:8082/
```

The command generates the following output:

```
HTTP/1.1 404 Not Found
Content-Length: 40
Content-Type: text/plain
```

No context-path matches the request URI.

i This response confirms that the Gateway has initialized, but no APIs have been deployed. Once APIs are published through the Control Plane, this message is replaced by valid responses routed through the configured context paths.

Verify the Redis Connection

- Test the Redis connection with the following command:

```
docker exec -it gio-apim-hybrid-redis redis-cli -a <redis_password> ping
```

The command generates the following output:

```
PONG
```

View the logs

- To check the Gateway logs, use the following command:

```
docker logs -f gio-apim-hybrid-gateway
```

The command generates the following output:

```
[gio-apim-hybrid-gateway] INFO i.g.p.c.internal.PluginRegistryImpl - Loading plugins from
[jio-apim-hybrid-gateway] INFO i.g.p.c.internal.PluginRegistryImpl - List of available plugins
[jio-apim-hybrid-gateway] INFO i.g.p.c.internal.PluginRegistryImpl - > rate-limiting
[jio-apim-hybrid-gateway] INFO i.g.p.c.internal.PluginRegistryImpl - > jwt [6.1.0]
[jio-apim-hybrid-gateway] INFO i.g.p.c.internal.PluginRegistryImpl - > key-less
[jio-apim-hybrid-gateway] INFO i.g.p.r.i.RepositoryPluginHandler - Repository [RATE_LIMITING]
[jio-apim-hybrid-gateway] INFO i.g.r.redis.vertx.RedisClient - Redis is now ready to be used
[jio-apim-hybrid-gateway] INFO i.g.node.container.AbstractContainer - Starting Gravitee.io
[jio-apim-hybrid-gateway] INFO i.g.g.r.s.vertx.HttpProtocolVerticle - HTTP server [http]
[jio-apim-hybrid-gateway] INFO i.g.node.container.AbstractNode - Gravitee.io - API Gateway
[jio-apim-hybrid-gateway] INFO i.g.g.s.s.p.r.s.n.NodeMetadataSynchronizer - Node metadata synchronized
[jio-apim-hybrid-gateway] INFO i.g.g.s.s.p.r.s.l.LicenseSynchronizer - 1 licenses synchronized
[jio-apim-hybrid-gateway] INFO i.g.g.s.s.p.r.s.api.ApiSynchronizer - 0 apis synchronized
[jio-apim-hybrid-gateway] INFO i.g.g.s.s.p.r.DefaultSyncManager - Sync service has been started
```

- To check the Redis logs, use the following command:

```
docker logs -f gio-apim-hybrid-redis
```

The command generates the following output:

```
1:C 19 Aug 2025 10:30:15.123 # 0000o000o000o Redis is starting o000o000o000o
1:C 19 Aug 2025 10:30:15.123 # Redis version=7.2.4, bits=64, commit=00000000, modified=0,
1:C 19 Aug 2025 10:30:15.123 # Configuration loaded
1:M 19 Aug 2025 10:30:15.124 * monotonic clock: POSIX clock_gettime
1:M 19 Aug 2025 10:30:15.124 * Running mode=standalone, port=6379.
1:M 19 Aug 2025 10:30:15.124 * Server initialized
1:M 19 Aug 2025 10:30:15.125 * Ready to accept connections tcp
```

(Optional) Stop the Gateway

To shut down the Gateway, choose one of the following options.

- This command stops the containers but keeps them available to restart. The Redis data is preserved.

```
docker stop gio-apim-hybrid-gateway gio-apim-hybrid-redis
```

- These commands remove the containers, their data, and the network, and all Redis data is lost.

```
docker stop gio-apim-hybrid-gateway gio-apim-hybrid-redis
docker rm gio-apim-hybrid-gateway gio-apim-hybrid-redis
# Remove the network
docker network rm gravitee-network
```

Proxy configuration

To route Gateway traffic through a corporate proxy (for example, for backend API calls or JWKS retrieval from external identity providers like Microsoft Entra ID), add the following `gravitee_system_proxy_*` environment variables to the Gateway container:

```
gravitee_system_proxy_enabled=true
gravitee_system_proxy_type=HTTP
gravitee_system_proxy_host=<proxy-host>
gravitee_system_proxy_port=<proxy-port>
gravitee_system_proxy_https_host=<proxy-host>
gravitee_system_proxy_https_port=<proxy-port>
```

For the full configuration reference including proxy authentication, see [System Proxy for Backend APIs](#). For an overview of all proxy methods, see [Proxy Configuration](#).

Next steps

- Access your API Management Console. To access your Console, complete the following steps:
 1. Log in to your [Gravitee Cloud ↗](#).
 2. From the Dashboard, navigate to the Environment where you created your Gateway.
 3. Click on **APIM Console** to open the user interface where you can create and manage your APIs.
- Create your first API. For more information about creating your first API, see [Create & Publish Your First API](#).
- Add native Kafka capabilities. For more information about adding native Kafka capabilities, see [Configure the Kafka Client & Gateway](#).